

$$sa) dS_t = (r - q) S_t dt + \sigma S_t dw_t$$

$$dv_t = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS_t + \frac{\partial^2 V}{\partial S^2} (dS_t)^2$$

$$X_t = \ln\left(\frac{S_t}{B_t}\right) = \ln(S_t) - \ln(B_t)$$

$$\frac{\partial X}{\partial S} = \frac{1}{S_t} \quad \frac{\partial^2 X}{\partial S^2} = \frac{-1}{S_t^2} \quad \frac{\partial X}{\partial B_t} = \frac{1}{B_t}$$

$$\begin{aligned} dX_t &= \frac{\partial X}{\partial B_t} dB_t + \frac{\partial X}{\partial S} dS_t + \frac{\partial^2 X}{\partial S^2} (dS_t)^2 \\ &= -r X_t dt + \frac{\partial X}{\partial S} dS_t + \frac{\partial^2 X}{\partial S^2} (dS_t)^2 \end{aligned}$$

$$dv_t + r X_t dt = \frac{\partial V}{\partial S} dS_t + \frac{\partial^2 V}{\partial S^2} (dS_t)^2$$

$$\frac{\partial V}{\partial S} = \frac{\partial V}{\partial X} \times \frac{\partial X}{\partial S} \quad \frac{\partial^2 V}{\partial S^2} = \frac{\partial^2 V}{\partial X^2} \cdot \frac{\partial X}{\partial S} + \frac{\partial V}{\partial X} \frac{\partial^2 X}{\partial S^2}$$

$$\hat{V}_t = \frac{V_t}{B_t}$$

$$\frac{\partial \hat{V}}{\partial B} = \frac{-V_t}{B_t^2} \quad \frac{\partial \hat{V}}{\partial V} = \frac{1}{B_t}$$

$$dv_t = \left(\frac{\partial V}{\partial t} - rV + (r - q)S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \frac{\partial V}{\partial S} \sigma S dw_t$$

$$d\hat{V}_t = \frac{\partial \hat{V}}{\partial B} dB_t + \frac{\partial \hat{V}}{\partial V} dV_t$$

$$= -\frac{rV_t B_t dt}{B_t^2} + \frac{1}{B_t} \left(\left(\frac{\partial V}{\partial t} + (r-q)S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV \right) dt + \frac{\partial V}{\partial S} \sigma S dw_t \right)$$

$$= \left(-\frac{rV_t B_t}{B_t^2} + \frac{B_t}{B_t^2} \left(\frac{\partial V}{\partial t} + (r-q)S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV \right) \right) dt + \frac{\sigma S \frac{\partial V}{\partial S} dw_t}{B_t}$$

$$= \frac{1}{B_t} \left(\frac{\partial V}{\partial t} + (r-q)S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - 2rV \right) dt + \frac{\sigma S \frac{\partial V}{\partial S} dw_t}{B_t}$$

$$= \frac{1}{B_t} \left(\frac{\partial V}{\partial t} + (r-q)S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - 2rB_t V \right) dt + \frac{\sigma S \frac{\partial V}{\partial S} dw_t}{B_t}$$

$$\frac{\partial V}{\partial S} = \frac{\partial V}{\partial X} \times \frac{\partial X}{\partial S} = \frac{1}{S_t} \frac{\partial V}{\partial X}$$

$$\frac{\partial^2 V}{\partial S^2} = \frac{\partial^2 V}{\partial X^2} \left(\frac{\partial X}{\partial S} \right)^2 + \frac{\partial V}{\partial X} \frac{\partial^2 X}{\partial S^2}$$

$$= \frac{1}{S_t^2} \frac{\partial^2 V}{\partial X^2} + \frac{\partial V}{\partial X} \frac{-1}{S_t^2}$$

$$= \frac{1}{S_t^2} \left(\frac{\partial^2 V}{\partial X^2} - \frac{\partial V}{\partial X} \right)$$

$$\therefore d\hat{V}_t = \frac{1}{B_t} \left(\frac{\partial V}{\partial t} + (r-q)S \frac{1}{S} \frac{\partial V}{\partial X} + \frac{1}{2} \sigma^2 \cancel{S^2} \frac{1}{S^2} \left(\frac{\partial^2 V}{\partial X^2} - \frac{\partial V}{\partial X} \right) - 2rV \right) dt + \frac{\sigma \cancel{S} \frac{1}{S} \frac{\partial V}{\partial X} dw_t}{B_t}$$

$$= \frac{1}{B_t} \left[\frac{\partial V}{\partial t} + (r-q) \frac{\partial V}{\partial X} + \frac{1}{2} \sigma^2 \left(\frac{\partial^2 V}{\partial X^2} - \frac{\partial V}{\partial X} \right) - 2rV \right] dt$$

$$+ \frac{\sigma}{B_t} \frac{\partial V}{\partial X} dW_t$$

$$= \frac{1}{B_t} \left[\frac{\partial V}{\partial t} + \left(r-q - \frac{1}{2} \sigma^2 \right) \frac{\partial V}{\partial X} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial X^2} - 2rB_t \hat{V} \right] dt$$

$$+ \frac{\sigma}{B_t} \frac{\partial V}{\partial X} dW_t$$

$$V = B_t \hat{V} \therefore \frac{\partial V}{\partial \hat{V}} = B_t$$

$$= \left[\left(\frac{\partial \hat{V}}{\partial \hat{V}} \times \frac{\partial V}{\partial t} \right) + \left(r-q - \frac{1}{2} \sigma^2 \right) \frac{\partial \hat{V}}{\partial \hat{V}} \frac{\partial V}{\partial X} + \frac{1}{2} \sigma^2 \frac{1}{B_t} \cdot B_t \frac{\partial^2 \hat{V}}{\partial X^2} - 2r \hat{V} \right] dt$$

$$+ \sigma \frac{\partial \hat{V}}{\partial \hat{V}} \frac{\partial V}{\partial X} dW_t$$

$$= \left(\frac{\partial \hat{V}}{\partial t} + \left(r-q - \frac{1}{2} \sigma^2 \right) \frac{\partial \hat{V}}{\partial X} + \frac{1}{2} \sigma^2 \frac{\partial^2 \hat{V}}{\partial X^2} - 2r \hat{V} \right) dt$$

$$+ \sigma \frac{\partial \hat{V}}{\partial X} dW_t$$

$$\text{PDE: } \frac{\partial \hat{V}}{\partial t} + \left(r-q - \frac{1}{2} \sigma^2 \right) \frac{\partial \hat{V}}{\partial X} + \frac{1}{2} \sigma^2 \frac{\partial^2 \hat{V}}{\partial X^2} - 2r \hat{V} = 0$$

$$b) \frac{\partial \hat{V}_{i,j}}{\partial t} \approx \frac{\hat{V}_{i,j+1} - \hat{V}_{i,j}}{\Delta t}$$

$$\frac{\partial \hat{V}}{\partial x} \approx \frac{\hat{V}_{i+1,j} - \hat{V}_{i-1,j}}{2\Delta x}$$

$$\frac{\partial^2 \hat{V}}{\partial x^2} \approx \frac{\hat{V}_{i+1,j} - 2\hat{V}_{i,j} + \hat{V}_{i-1,j}}{(\Delta x)^2}$$

$$\left[\frac{\hat{V}_{i,j+1} - \hat{V}_{i,j}}{\Delta t} + (r - q + \frac{1}{2}\sigma^2) \frac{\hat{V}_{i+1,j} - \hat{V}_{i-1,j}}{2\Delta x} + \frac{1}{2}\sigma^2 \frac{\hat{V}_{i+1,j} - 2\hat{V}_{i,j} + \hat{V}_{i-1,j}}{(\Delta x)^2} \right] = 0$$

c. Numerically more stable, one grid search and computationally faster.

$$6a) \Delta X_t = \theta(\mu - x_i) \Delta t + \sigma \sqrt{\Delta t} z_i$$

$$\Delta X_t = X_{i+1} - X_i \quad (\text{in this scenario})$$

$$\therefore X_{i+1} = X_i + \theta(\mu - x_i) \Delta t + \sigma \sqrt{\Delta t} z_i$$

$$b. \quad dX_t = \theta \mu dt - \theta X_t dt + \sigma dW_t$$

$$\hat{V} = \frac{V_t}{B_t} \quad V_t = \hat{V} B_t$$

$$\frac{\partial \hat{V}}{\partial B} = -\frac{V_t}{B_t^2}$$

$$\frac{\partial \hat{V}}{\partial V} = \frac{1}{B_t}$$

$$dV_t = \frac{\partial V}{\partial X} dX_t + \frac{1}{2} \frac{\partial^2 V}{\partial X^2} (dX_t)^2$$

$$d\hat{V}_t = \frac{\partial \hat{V}}{\partial B_t} dt + \frac{\partial \hat{V}}{\partial V} dV_t$$

$$= -r\hat{V}dt + \frac{\partial \hat{V}}{\partial V} \left(\frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial X} (\theta(\mu - X_t)dt + \sigma dW_t) + \frac{1}{2} \frac{\partial^2 V}{\partial X^2} \sigma^2 dt \right)$$

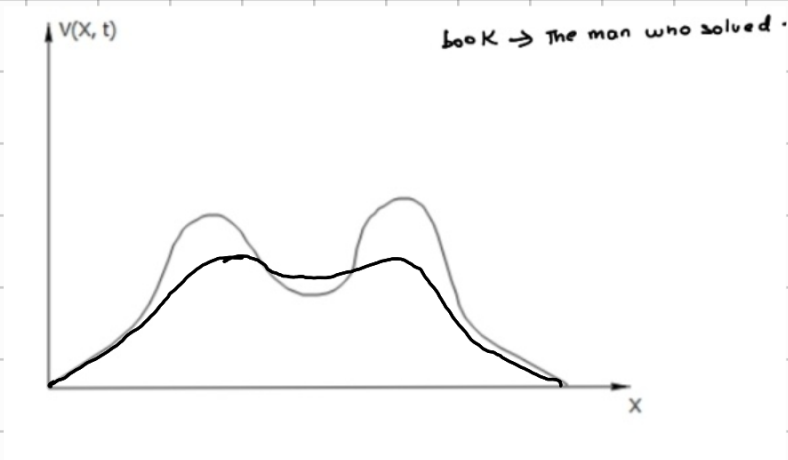
$$= \frac{\partial \hat{V}}{\partial t} dt - r\hat{V}dt + \frac{\partial \hat{V}}{\partial X} (\theta(\mu - X_t)dt) + \frac{\partial \hat{V}}{\partial X} \sigma dW_t + \frac{1}{2} \frac{\partial^2 \hat{V}}{\partial X^2} \sigma^2 dt$$

$$= \left(\frac{\partial \hat{V}}{\partial t} + \theta(\mu - X_t) \frac{\partial \hat{V}}{\partial X} + \frac{\sigma^2}{2} \frac{\partial^2 \hat{V}}{\partial X^2} - r\hat{V} \right) dt + \frac{\partial \hat{V}}{\partial X} \sigma dW_t$$

$$\text{PDE: } \frac{\partial \hat{V}}{\partial t} + \theta(\mu - x_t) \frac{\partial \hat{V}}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 \hat{V}}{\partial x^2} - r\hat{V} = 0$$

$$c. \left[\frac{\hat{V}_{i,j} - \hat{V}_{i,j-1}}{\Delta t} + \theta(\mu - x_t) \frac{V_{i+1,j} + V_{i-1,j}}{2\Delta x} + \frac{\sigma^2}{2} \frac{V_{i+1,j} - 2V_{i,j} + V_{i-1,j}}{(\Delta x)^2} - rV_{i,j} \right] = 0$$

d.



It will be smoother due to the mean reverting process of the model. So as time passes ^{to 0 (r=0)} it mean reverts, but at a slow pace as $\theta = 0$.

$$S(a) \quad X_t = \ln\left(\frac{S_t}{B_t}\right) = \ln(S_t) - \ln(B_t)$$

$$\frac{\partial X}{\partial B} = -\frac{1}{B_t}$$

$$\frac{\partial X}{\partial S} = \frac{1}{S_t}$$

$$\frac{\partial^2 X}{\partial S^2} = -\frac{1}{S_t^2}$$

$$\begin{aligned} dx_t &= \frac{\partial X}{\partial B} dB_t + \frac{\partial X}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 X}{\partial S^2} (dS_t)^2 \\ &= -\frac{1}{B_t} r B_t dt + \frac{1}{S_t} ((r-q)dt + \sigma dW_t) - \frac{1}{2} \frac{1}{S_t^2} \sigma^2 S_t^2 dt \\ &= -\cancel{r} dt + \cancel{r} dt - q dt + \sigma dW_t - \frac{1}{2} \sigma^2 dt \\ &= \left(-q - \frac{1}{2} \sigma^2\right) dt + \sigma dW_t \end{aligned}$$

$$\begin{aligned} d\hat{V}_t &= \frac{\partial \hat{V}}{\partial t} dt + \frac{\partial \hat{V}}{\partial X} dx_t + \frac{1}{2} \frac{\partial^2 \hat{V}}{\partial X^2} (dx_t)^2 \\ &= \frac{\partial \hat{V}}{\partial t} dt + \frac{\partial \hat{V}}{\partial X} \left(\left(-q - \frac{1}{2} \sigma^2\right) dt + \sigma dW_t \right) + \frac{1}{2} \frac{\partial^2 \hat{V}}{\partial X^2} \sigma^2 dt \\ &= \left(\frac{\partial \hat{V}}{\partial t} + \left(-q - \frac{1}{2} \sigma^2\right) \frac{\partial \hat{V}}{\partial X} + \frac{1}{2} \sigma^2 \frac{\partial^2 \hat{V}}{\partial X^2} \right) dt + \sigma \frac{\partial \hat{V}}{\partial X} dW_t \end{aligned}$$

POE:

$$\frac{\partial \hat{V}}{\partial t} + \left(-q - \frac{1}{2} \sigma^2\right) \frac{\partial \hat{V}}{\partial X} + \frac{1}{2} \sigma^2 \frac{\partial^2 \hat{V}}{\partial X^2} = 0$$

6b. $\hat{V}_t = \frac{V_t}{B_t}$ $V_t = f(x_t, t)$

$\frac{\partial \hat{V}}{\partial B} = -\frac{V_t}{B_t^2}$ ← Properly diff

$\frac{\partial \hat{V}}{\partial V} = \frac{1}{B_t}$

$dV_t = \frac{\partial \hat{V}}{\partial B} dB_t + \frac{\partial \hat{V}}{\partial V} dV_t$

$= -\frac{rV_t}{B_t} dt + \frac{1}{B_t} \left(\frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial x} dx_t + \frac{1}{2} \frac{\partial^2 V}{\partial x^2} dx_t^2 \right)$

$= \frac{1}{B_t} \left(\frac{\partial V}{\partial t} + \theta(\mu - x_t) \frac{\partial V}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial x^2} - rV \right) dt + \frac{\sigma \partial V}{B_t \partial x} dw_t$

∴ PDE:

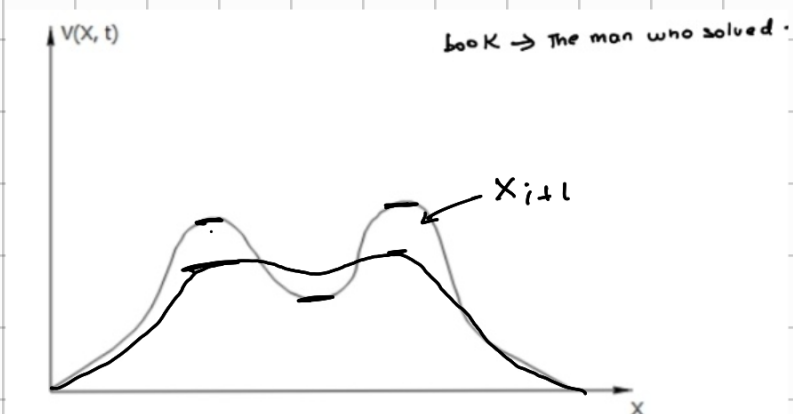
$\frac{\partial V}{\partial t} + \theta(\mu - x_t) \frac{\partial V}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial x^2} - rV = 0$

formula: $\theta(\mu - x_t)$ and $-rV \rightarrow$ part c

$$\Delta x = x_{i+1} - x_i$$

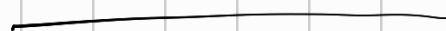
$$x_{i+1} = x_i + \dots$$

$\theta = 0 \rightarrow$ mean reversion $\rightarrow 0$
and no drift
(slow slope and smooth)



$$-r = 0$$

$$\text{mean} = 0$$



$$4a. \mathbb{E}[f(x)] = \frac{1}{b-a} \int_a^b f(x) dx$$

$$\mathbb{E}[f(x)] = \frac{1}{2-0} \int_0^2 f(x) dx$$

$$\therefore \int_0^2 f(x) dx = 2 \mathbb{E}[f(x)]$$

Algo 4: $\hat{\mu}_n = MC(n)$

$s = 0$

for i in 1 to n do

Generate $\vec{x}_i \sim U(0, 2)$

$$h_i = h(\vec{x}_i)$$

$$s += h_i$$

end for

$$\hat{\mu}_n = 2 * [s/n]$$

return $\hat{\mu}_n$

$$b) g(x) = f(0) + f'(0)(x-0) + \frac{1}{2}f''(0)(x-0)^2$$

$$f(0) = \frac{1}{2}e^{-\frac{1}{2}(0)^2} = \frac{1}{2}$$

$$f'(x) = \frac{1}{2}e^{-\frac{1}{2}x^2} \times \frac{d(\frac{1}{2}x^2)}{dx} = -x \frac{1}{2}e^{-\frac{1}{2}x^2}$$

$$f'(0) = 0$$

$$f''(x) = \frac{d(-x)}{dx} \times \frac{1}{2}e^{-\frac{1}{2}x^2} + \frac{d(\frac{1}{2}e^{-\frac{1}{2}x^2})}{dx} \times (-x)$$

$$= -\frac{1}{2}e^{-\frac{1}{2}x^2} + \frac{x^2}{2}e^{-\frac{1}{2}x^2}$$

Remember to check your deriv

$$f''(0) = -\frac{1}{2}$$

$$\therefore g(x) = \frac{1}{2} + 0(x) + \frac{1}{2}\left(-\frac{1}{2}\right)x^2$$

$$= \frac{1}{2} - \frac{1}{4}x^2$$

$$g^* = \int_0^2 g(x) dx$$

$$= \int_0^2 \frac{1}{2} - \frac{1}{4}x^2 dx$$

$$= \int_0^2 \frac{1}{2} dx - \int_0^2 \frac{1}{4}x^2 dx$$

$$= \left[\frac{x}{2} \right]_0^2 - \left[\frac{x^3}{12} \right]_0^2$$

$$= \left[\frac{2}{2} - \frac{0}{2} \right] - \left[\frac{2^3}{12} - \frac{0^3}{12} \right]$$

$$= [1 - 0] - \left[\frac{2}{3} \right] = 1 - \frac{2}{3} = \frac{3-2}{3} = \frac{1}{3}$$

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c. def mcIntegrate(f, a, b, g, g-integral):
    S = 0
    for i = 1 to n do:
        x_i = np.random.uniform(a, b) # Generate  $\vec{x}_i \sim U(a, b)$ 
        h_i = f(x_i)
        g_mean = g-integral
        g_i = g(x_i)
        S += h_i + beta(g_mean - g_i)
        # beta calculated separately using simulation.
    end for
     $\hat{\mu}_n = S/n$ 
    return  $\hat{\mu}_n$ 

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